

Isomorphism Theorems

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1 Group

Theorem 1.1. (*First Isomorphism Theorem for Groups*)

Let $\varphi : G \rightarrow H$ be a group homomorphism. Then,

$$\ker \varphi \trianglelefteq G \quad , \quad G / \ker \varphi \cong \varphi(G)$$

Theorem 1.2. (*Second/Diamond Isomorphism Theorem for Groups*)

Let $A, B \leq G$, assume $A \leq N_G(B)$. Then,

1. $AB \leq G$
2. $B \trianglelefteq AB$
3. $A \cap B \trianglelefteq A$
4. $AB/B \cong A/A \cap B$

Theorem 1.3. (*Third Isomorphism Theorem for Groups*)

Let $H, K \trianglelefteq G$ with $H \leq K$. Then,

1. $K/H \trianglelefteq G/H$
2. $(G/H)/(K/H) \cong G/K$

Theorem 1.4. (*Fourth/Lattice Isomorphism Theorem for Groups*)

Let $N \trianglelefteq G$.

Construct $S := \{A \in \mathcal{P}(G) \mid N \subseteq A \leq G\}$, $S_q := \{Y \in \mathcal{P}(G/N) \mid Y \leq G/N\}$.

We denote $\overline{A} := A/N$ (left cosets), for $A \in S$.

Define map $\varphi : S \rightarrow S_q$ as,

$$A \mapsto A/N$$

Then, φ is a bijection.

Further, $\forall A, B \in S$ we have

1. $A \leq B \iff \overline{A} \leq \overline{B}$
2. If $A \leq B$, then $|B : A| = |\overline{B} : \overline{A}|$
3. $\overline{\langle A, B \rangle} = \langle \overline{A}, \overline{B} \rangle$
4. $\overline{A \cap B} = \overline{A} \cap \overline{B}$
5. $A \trianglelefteq G \iff \overline{A} \trianglelefteq \overline{G}$

2 Ring

Theorem 2.1. (First Isomorphism Theorem for Rings)

Let $\varphi : R \rightarrow S$ be a ring homomorphism. Then, $R/\ker \varphi \cong \varphi(R)$

Proof. Note, φ is a group homomorphism between $(R, +_R)$ and $(S, +_S)$.

By first isomorphism theorem for groups, we know of a bijection $\Phi : R/\ker(\varphi) \rightarrow \varphi(R)$,

$$r + \ker(\varphi) \mapsto \varphi(r)$$

We know $\varphi(R)$ is a subring of S , as the image of a ring homomorphism, so it can be viewed as an independent ring with induced operations.

We also know $\ker \varphi$ is an ideal of R , as the kernel of a ring homomorphism, so the left cosets $R/\ker(\varphi)$ has the usual quotient ring structure.

From group, we know Φ is an bijection and is additive. So, it **S.T.S.** multiplicativity.

Take any $r + \ker(\varphi), s + \ker(\varphi) \in R/\ker(\varphi)$, we have

$$\Phi((r + \ker(\varphi)) \cdot_{R/\ker(\varphi)} (s + \ker(\varphi))) = \Phi((r \cdot_R s) + \ker(\varphi)) = \varphi(r \cdot_R s) = \varphi(r) \cdot_S \varphi(s)$$

□

Theorem 2.2. (Second Isomorphism Theorem for Rings)

Let R be a ring. Let A be a subring of R , B an ideal of R . Then,

1. $A + B = \{a + b \in R \mid a \in A, b \in B\}$ is a subring of R
2. $A \cap B$ is an ideal of A
3. $(A + B)/B \cong A/(A \cap B)$

Proof. We first check conditions of second isomorphism theorem for groups. Consider the additive group $(R, +)$, $A, B \leq (R, +)$, and B is a normal subgroup for $(R, +)$ is abelian.

By group, we know $A + B \leq (R, +)$. Take any $a + b, a' + b' \in A + B$,

$$(a + b)(a' + b') = aa' + ba' + ab' + bb' \in A + B$$

because B is an ideal of R , and A is a subring (closed under multiplication). So, $A + B$ is a subring of R .

Note, $B \subseteq A + B$, since both are subrings, we have B is a subring of $A + B$ (considered as ring with induced operations).

Take any $b \in B, x \in A + B$, we have

$$b \cdot_{A+B} x = bx \in B, \quad x \cdot_{A+B} b = xb \in B$$

So, B is an ideal of $A + B$.

As such, $(A + B)/B$ is well-defined as a quotient ring.

Now, we apply second isomorphism theorem for groups. We have $\varphi : A \rightarrow (A + B)/B$ defined as,

$$a \mapsto a + B$$

We knew from group, that φ is a surjective group homomorphism from $(A, +_A)$ to $((A + B)/B, +_{(A+B)/B})$, and that $\ker(\varphi) = A \cap B$

Take $a, a' \in A$, we have

$$\varphi(aa') = (aa') + B = (a + B) \cdot_{(A+B)/B} (a' + B) = \varphi(a) \cdot_{(A+B)/B} \varphi(a')$$

So, φ is a surjective ring homomorphism. By first isomorphism theorem for rings above, we have

$$A/\ker(\varphi) = A/(A \cap B) \cong (A + B)/B$$

And $A \cap B$ is an ideal of A . □

Theorem 2.3. (Third Isomorphism Theorem for Rings)

Let R be a ring. Let I, J be ideals of R , $I \subset J$. Then,

1. J/I is an ideal of R/I
2. $(R/I)/(J/I) \cong R/J$

Proof. Observe I is an ideal of J (considered as a ring).

From group, we have $\varphi : R/I \rightarrow R/J$ as,

$$r + I \mapsto r + J$$

Recall this φ is a surjective group homomorphism, and that $\ker(\varphi) = J/I$.

Take $r + I, s + I \in R/I$, we have

$$\varphi(rs + I) = rs + J = (r + J)(s + J) = \varphi(r + I)\varphi(s + I)$$

Thus, φ is a surjective ring homomorphism.

By first isomorphism theorem for rings, we know

$$(R/I)/(J/I) \cong R/J \quad \text{and} \quad J/I \text{ is an ideal of } R/I$$

□

Theorem 2.4. (Fourth Isomorphism Theorem for Rings)

Let R be a ring. Let I be an ideal of R .

Construct T as the collection of all subrings of R that contains I .

Construct T_q as the collection of all subrings of R/I . Then,

1. $A \mapsto A/I : T \rightarrow T_q$ is a bijection
2. Let $A \in T$. Then, A is an ideal of $R \iff A/I$ is an ideal of R/I

Proof. Note, $I \trianglelefteq (R, +)$. And recall our constructions from group,

1. $S := \{A \in \mathcal{P}(R) \mid I \subseteq A \leq (R, +)\}$
2. $S_q := \{Y \in \mathcal{P}(R/I) \mid Y \leq (R/I, +_{R/I})\}$
3. Bijection $\varphi : S \rightarrow S_q$,

$$A \mapsto A/I$$

Observe $T \subseteq S$ and $T_q \subseteq S_q$. Consider restriction of φ , we define $\psi : T \rightarrow T_q$ as,

$$A \mapsto \varphi(A)$$

Note, we must show that $\varphi(A) \in T_q$, for all $A \in T$.

Take any $a + I, a' + I \in A/I$, we have

$$(a + I)(a' + I) = aa' + I \in A/I$$

because $aa' \in A$ as A is a subring of R . So, ψ is injective and well-defined.

Now, take any $Y \in T_q$, then $Y \in S_q$, $\exists A \in S$ s.t. $\varphi(A) = Y = A/I$.

Take any $a_1, a_2 \in A$, observe $a_i + I \in Y \implies (a_1 a_2) + I \in Y = A/I$.

$\implies \exists a_0 \in A$ s.t. $a_0 + I = (a_1 a_2) + I \implies a_1 a_2 \in A$, as $I \subseteq A$.

Thus, A is a subring of R , and $A \in T$.

Hence, ψ is a bijection.

($\implies$) Suppose A is an ideal of R . We know by ψ , A/I is a subring of R/I .

Take any $a + I \in A/I$, $r + I \in R/I$, we have

$$(a + I)(r + I) = ar + I \in A + I$$

Similar for the other direction.

($\impliedby$) Suppose A/I is an ideal of R/I . Take any $a \in A$, $r \in R$, we have

$$(a + I)(r + I) = ar + I \in A/I$$

But since $I \subseteq A$, we have $ar \in A$. Similar, for the other direction. □

3 Module

Theorem 3.1. (First Isomorphism Theorem for Modules)

Let M, N be R -modules, $\varphi \in \text{Hom}_R(M, N)$.

Then, $\ker \varphi$ is a submodule of M , and $M/\ker \varphi \cong \varphi(M)$.

Proof. Observe φ is a group homomorphism between the additive groups of M, N . So, we apply first isomorphism theorem for groups, and obtain:

1. $\ker(\varphi) \leq (M, +_M)$

2. Group isomorphism $\Phi : M/\ker \varphi \rightarrow \varphi(M)$,

$$m + \ker \varphi \mapsto \varphi(m)$$

Take any $r \in R$, $x \in \ker \varphi$, we have $\varphi(rx) = r\varphi(x) = r0_N = 0_N$.

So, $\ker \varphi$ is a submodule.

Take any $r \in R$, $m + \ker \varphi \in M/\ker \varphi$, we have

$$\Phi(r(m + \ker \varphi)) = \Phi((rm) + \ker \varphi) = \varphi(rm) = r\varphi(m) = r\Phi(m + \ker \varphi)$$

We are done. □

Theorem 3.2. (Second Isomorphism Theorem for Modules)

Let A, B be submodules of R -module M . Then,

$$(A + B)/B \cong A/(A \cap B)$$

Proof. Note, $B \subseteq A + B$, and when $A + B$ is viewed as a R -module, $B \leq_{\text{module}} A + B$. (proof obvious)

Notice $(M, +)$ is an additive group, $A, B, A + B, A \cap B \leq (M, +)$.

Meanwhile, they are also submodules of module M .

Apply second isomorphism theorem for groups, we obtain map $\varphi : A \rightarrow (A + B)/B$,

$$a \mapsto a + B$$

Recall φ is a surjective group homomorphism, and $\ker \varphi = A \cap B$.

Now, observe both $A, (A + B)/B$ have quotient module structure defined, in which their additive groups matches the isomorphism theorem for groups' condition. So, it suffices for us to check that φ is R -homogeneous. If holds, then φ is a surjective R -module homomorphism. Then, by first isomorphism theorem for modules, we can conclude.

Take any $r \in R$, $a \in A$, we have

$$\varphi(ra) = (ra) + B = r(a + B) = r\varphi(a)$$

We are done. □

Theorem 3.3. (Third Isomorphism Theorem for Modules)

Let A, B be submodules of R -module M , and $A \subset B$. Then,

$$(M/A)/(B/A) \cong M/B$$

Proof. Observe the conditions for invoking third isomorphism theorem for groups are present. We obtain, A surjective group homomorphism with kernel B/A (set theoretically, left cosets), $\varphi : M/A \rightarrow M/B$,

$$m + A \mapsto m + B$$

Once again, it suffices to show that φ is R -homogeneous.

Take any $r \in R$, $m + A \in M/A$, we have

$$\varphi(r(m + A)) = \varphi((rm) + A) = (rm) + B = r(m + B) = r\varphi(m)$$

□

Theorem 3.4. (Fourth Isomorphism Theorem for Modules)

Let N be a submodule of R -module M .

Construct L as the collection of all submodules of M that contains N .

Construct L_q as the collection of all submodules of M/N .

Then, $A \mapsto A/N : L \rightarrow L_q$ is an bijection.

Further, this correspondence commutes with the processes of taking sums and intersections, i.e.

1. $A_1/N \cap A_2/N = (A_1 \cap A_2)/N$
2. $A_1/N + A_2/N = (A_1 + A_2)/N$

Proof. We will make heavy use of fourth isomorphism theorem for groups and notations in its proof. Observe $N \trianglelefteq (M, +)$, we construct

1. $S := \{A \in \mathcal{P}(M) \mid N \subseteq A \leq (M, +)\}$
2. $S_q := \{Y \in \mathcal{P}(M/N) \mid Y \leq (M/N, +_{M/N})\}$
3. Bijection $\varphi : S \rightarrow S_q$,

$$A \mapsto A/N$$

Observe $L \subseteq S$ and $L_q \subseteq S_q$. Consider restriction of φ , we define $\psi : L \rightarrow L_q$ as,

$$A \mapsto \varphi(A)$$

Note, we must show that $\varphi(A) \in L_q$, for all $A \in L$.

Take any $r \in R$, $a + N \in A/N$, we have

$$r(a + N) = (ra) + N \in A/N$$

So, $A/N \in L_q$. So, codomain of ψ is justified. To show ψ is bijective, we only need to show surjectivity.

Take any $Y \in L_q \subseteq S_q \implies \exists A \in S$ s.t. $\varphi(A) = A/N = Y$.

Take any $r \in R$, $a \in A$, we have

$$r(a + N) \in A/N \implies \exists(a' + N \in A/N) \ r(a + N) = (ra) + N = a' + N \implies ra \in A$$

Thus, $A \in L$. Hence, ψ is a bijection.

Recall from group, we have that $\forall A, B \in S$, $\varphi(A \cap B) = \varphi(A) \cap \varphi(B)$.
 Now, take any $A_1, A_2 \in L$, we have

$$\psi(A_1) \cap \psi(A_2) = \varphi(A_1) \cap \varphi(A_2) = \varphi(A_1 \cap A_2) = \psi(A_1 \cap A_2)$$

Notice, $A \cap B \in L$, so $\psi(A \cap B)$ is valid.

Recall from group, we have that $\forall A, B \in S$, $\varphi(\langle A, B \rangle_{(M, +_M)}) = \langle \varphi(A), \varphi(B) \rangle_{(M/N, +_{M/N})}$.

Well, in abelian additive group, join of two subgroups A, B are precisely $A + B$.

Now, take any $A_1, A_2 \in L$, we have

$$\psi(A_1) + \psi(A_2) = \varphi(A_1) + \varphi(A_2) = \langle \varphi(A_1), \varphi(A_2) \rangle_{(M/N, +_{M/N})} = \varphi(\langle A_1, A_2 \rangle_{(M, +_M)}) = \varphi(A_1 + A_2) = \psi(A_1 + A_2)$$

□

4 Factor Through by Substructure

4.1 Group

Lemma 4.1. (*Factor group homomorphism through normal subgroups*)

Let G, H be groups, $N \trianglelefteq G$.

Let $\varphi : G \rightarrow H$ be a group homomorphism.

We define $\Phi : G/N \rightarrow H$ as,

$$gN \mapsto \varphi(g)$$

Then, Φ is well-defined $\iff N \subseteq \ker \varphi$.

Further, if the above holds, then Φ is a group homomorphism.

4.2 Ring

Lemma 4.2. (*Factor ring homomorphism through ideals*)

Let R, S be rings, $I \trianglelefteq_{\text{ring}} R$.

Let $\varphi : R \rightarrow S$ be a ring homomorphism.

We define $\Phi : R/I \rightarrow S$ as,

$$r + I \mapsto \varphi(r)$$

Then, Φ is well-defined $\iff I \subseteq \ker \varphi$.

Further, if the above holds, then Φ is a ring homomorphism.

Notice, we don't need to do anything, just apply group version. Then, verify the last line.

4.3 Module

Lemma 4.3. (*Factor module homomorphism through submodules*)

Let R be a ring, M, M' be R -modules, $N \leq_{\text{module}} M$.

Let $\varphi : M \rightarrow M'$ be an R -module homomorphism.

We define $\Phi : M/N \rightarrow M'$ as,

$$m + N \mapsto \varphi(m)$$

Then, Φ is well-defined $\iff N \subseteq \ker \varphi$.

Further, if the above holds, then Φ is an R -module homomorphism.

Similar to ring case.

5 Transport of Structure

5.1 Group

Theorem 5.1. (*Transport of group structure*)

Let $(G, *)$ be a group, let H be a set, and let

$$\sigma : G \rightarrow H$$

be a bijection.

Define a binary operation $\circ$ on H by

$$x \circ y := \sigma(\sigma^{-1}(x) * \sigma^{-1}(y)), \quad \forall x, y \in H.$$

Then $(H, \circ)$ is a group. Moreover,

$$\sigma : (G, *) \rightarrow (H, \circ)$$

is a group isomorphism.

Proof. We prove that $(H, \circ)$ satisfies the group axioms.

(associativity) Take any $x, y, z \in H$. Then

$$(x \circ y) \circ z = \sigma(\sigma^{-1}(x \circ y) * \sigma^{-1}(z)).$$

By definition of $\circ$,

$$x \circ y = \sigma(\sigma^{-1}(x) * \sigma^{-1}(y)),$$

hence

$$\sigma^{-1}(x \circ y) = \sigma^{-1}(x) * \sigma^{-1}(y).$$

Therefore

$$(x \circ y) \circ z = \sigma((\sigma^{-1}(x) * \sigma^{-1}(y)) * \sigma^{-1}(z)).$$

Similarly,

$$x \circ (y \circ z) = \sigma(\sigma^{-1}(x) * (\sigma^{-1}(y) * \sigma^{-1}(z))).$$

Since $*$ is associative on G , it follows that

$$(x \circ y) \circ z = x \circ (y \circ z).$$

(identity) Let 1_G be the identity element of G . Define

$$1_H := \sigma(1_G) \in H.$$

We claim that 1_H is the identity element of H .

Take any $x \in H$. Then

$$1_H \circ x = \sigma(\sigma^{-1}(1_H) * \sigma^{-1}(x)) = \sigma(1_G * \sigma^{-1}(x)) = \sigma(\sigma^{-1}(x)) = x.$$

Similarly,

$$x \circ 1_H = \sigma(\sigma^{-1}(x) * \sigma^{-1}(1_H)) = \sigma(\sigma^{-1}(x) * 1_G) = \sigma(\sigma^{-1}(x)) = x.$$

Thus 1_H is an identity element of H .

(inverses) Take any $x \in H$. Define

$$x_H^{-1} := \sigma((\sigma^{-1}(x))_G^{-1}).$$

We claim that x_H^{-1} is an inverse of x in H .

Indeed,

$$x \circ x_H^{-1} = \sigma(\sigma^{-1}(x) * \sigma^{-1}(x_H^{-1})) = \sigma(\sigma^{-1}(x) * (\sigma^{-1}(x))_G^{-1}) = \sigma(1_G) = 1_H.$$

Similarly,

$$x_H^{-1} \circ x = 1_H.$$

Hence every element of H has an inverse.

Therefore $(H, \circ)$ is a group.

Finally, let $a, b \in G$. Then

$$\sigma(a) \circ \sigma(b) = \sigma(\sigma^{-1}(\sigma(a)) * \sigma^{-1}(\sigma(b))) = \sigma(a * b).$$

Thus

$$\sigma(a * b) = \sigma(a) \circ \sigma(b),$$

so σ is a group homomorphism. Since σ is bijective, it is a group isomorphism. $\square$

Remark: The group structure on H is the unique group structure for which σ becomes a group isomorphism. This is called the transport of group structure along the bijection σ . To be precise, we have:

Corollary 5.1.1. (Uniqueness of group structure permitting isomorphism)

Let $(G, *)$ be a group, let H be a set, and let

$$\sigma : G \rightarrow H$$

be a bijection. Then there exists a **unique** binary operation $\circ$ on H such that

$$\sigma : (G, *) \rightarrow (H, \circ)$$

is a group isomorphism.

Proof. **(existence)** By previous theorem.

(uniqueness) Suppose $\diamond$ is another binary operation on H such that

$$\sigma : (G, *) \rightarrow (H, \diamond)$$

is a group isomorphism.

Take any $x, y \in H$. Since σ is bijective, there exist unique $a, b \in G$ such that

$$x = \sigma(a), \quad y = \sigma(b).$$

Because σ is a homomorphism with respect to $\diamond$, we have

$$x \diamond y = \sigma(a) \diamond \sigma(b) = \sigma(a * b).$$

On the other hand, by the definition of $\circ$,

$$x \circ y = \sigma(\sigma^{-1}(x) * \sigma^{-1}(y)) = \sigma(a * b).$$

Hence

$$x \diamond y = x \circ y.$$

Since $x, y \in H$ were arbitrary, it follows that

$$\diamond = \circ.$$

Therefore the binary operation $\circ$ is unique. $\square$

5.2 Ring

Theorem 5.2. (*Transport of ring structure and uniqueness*)

Let $(R, +, \cdot)$ be a ring, let S be a set, and let

$$\sigma : R \rightarrow S$$

be a bijection.

Define binary operations $\oplus$ and $\otimes$ on S by

$$x \oplus y := \sigma(\sigma^{-1}(x) + \sigma^{-1}(y)),$$

$$x \otimes y := \sigma(\sigma^{-1}(x) \cdot \sigma^{-1}(y)), \quad \forall x, y \in S.$$

Then $(S, \oplus, \otimes)$ is a ring, and

$$\sigma : (R, +, \cdot) \rightarrow (S, \oplus, \otimes)$$

is a ring isomorphism.

Moreover, $(\oplus, \otimes)$ is the unique ring structure on S for which σ is a ring isomorphism.

Proof. We first prove existence.

Consider the additive group $(R, +)$. Since $\sigma : R \rightarrow S$ is a bijection, the group transport theorem shows that the operation

$$x \oplus y := \sigma(\sigma^{-1}(x) + \sigma^{-1}(y))$$

makes $(S, \oplus)$ into an abelian group, and that

$$\sigma : (R, +) \rightarrow (S, \oplus)$$

is a group isomorphism.

Now define multiplication on S by

$$x \otimes y := \sigma(\sigma^{-1}(x) \cdot \sigma^{-1}(y)).$$

We prove that $(S, \oplus, \otimes)$ is a ring.

(associativity of multiplication) Take any $x, y, z \in S$. Then

$$(x \otimes y) \otimes z = \sigma(\sigma^{-1}(x \otimes y) \cdot \sigma^{-1}(z)).$$

By definition of $\otimes$,

$$x \otimes y = \sigma(\sigma^{-1}(x) \cdot \sigma^{-1}(y)),$$

hence

$$\sigma^{-1}(x \otimes y) = \sigma^{-1}(x) \cdot \sigma^{-1}(y).$$

Therefore

$$(x \otimes y) \otimes z = \sigma((\sigma^{-1}(x) \cdot \sigma^{-1}(y)) \cdot \sigma^{-1}(z)).$$

Similarly,

$$x \otimes (y \otimes z) = \sigma(\sigma^{-1}(x) \cdot (\sigma^{-1}(y) \cdot \sigma^{-1}(z))).$$

Since multiplication on R is associative, it follows that

$$(x \otimes y) \otimes z = x \otimes (y \otimes z).$$

(distributive laws) Take any $x, y, z \in S$. Then

$$(x \oplus y) \otimes z = \sigma(\sigma^{-1}(x \oplus y) \cdot \sigma^{-1}(z)).$$

Since $\oplus$ was defined by transport of the additive structure,

$$\sigma^{-1}(x \oplus y) = \sigma^{-1}(x) + \sigma^{-1}(y).$$

Hence

$$(x \oplus y) \otimes z = \sigma\left((\sigma^{-1}(x) + \sigma^{-1}(y)) \cdot \sigma^{-1}(z)\right).$$

Using distributivity in R ,

$$= \sigma\left(\sigma^{-1}(x) \cdot \sigma^{-1}(z) + \sigma^{-1}(y) \cdot \sigma^{-1}(z)\right).$$

Since σ is additive with respect to $+$ and $\oplus$,

$$(x \oplus y) \otimes z = \sigma(\sigma^{-1}(x) \cdot \sigma^{-1}(z)) \oplus \sigma(\sigma^{-1}(y) \cdot \sigma^{-1}(z)) = (x \otimes z) \oplus (y \otimes z).$$

The proof of

$$x \otimes (y \oplus z) = (x \otimes y) \oplus (x \otimes z)$$

is similar.

Therefore $(S, \oplus, \otimes)$ is a ring.

Now we show that σ is a ring homomorphism. Since $\oplus$ was obtained by transporting the additive group structure, σ is already an isomorphism of additive groups. It remains to check compatibility with multiplication. Take any $a, b \in R$. Then

$$\sigma(a) \otimes \sigma(b) = \sigma(\sigma^{-1}(\sigma(a)) \cdot \sigma^{-1}(\sigma(b))) = \sigma(a \cdot b).$$

Thus

$$\sigma(a \cdot b) = \sigma(a) \otimes \sigma(b).$$

So σ is a ring isomorphism.

This proves existence.

Now we prove uniqueness.

Suppose $(\oplus', \otimes')$ is another ring structure on S such that

$$\sigma : (R, +, \cdot) \rightarrow (S, \oplus', \otimes')$$

is a ring isomorphism.

Since σ is then, in particular, an isomorphism of additive groups, the uniqueness statement in the group transport theorem implies that $\oplus' = \oplus$.

It remains to show that $\otimes' = \otimes$. Take any $x, y \in S$. Since σ is bijective, there exist unique $a, b \in R$ such that

$$x = \sigma(a), \quad y = \sigma(b).$$

Because σ is multiplicative with respect to $\otimes'$,

$$x \otimes' y = \sigma(a) \otimes' \sigma(b) = \sigma(a \cdot b).$$

On the other hand, by definition of $\otimes$,

$$x \otimes y = \sigma(\sigma^{-1}(x) \cdot \sigma^{-1}(y)) = \sigma(a \cdot b).$$

Hence

$$x \otimes' y = x \otimes y.$$

Since $x, y \in S$ were arbitrary, it follows that

$$\otimes' = \otimes.$$

Therefore the ring structure $(\oplus, \otimes)$ on S is unique. □

If R has additional properties, they are automatically transported too.

For example:

1. if R is commutative, then $(S, \oplus, \otimes)$ is commutative;
2. if R has identity 1_R , then

$$1_S := \sigma(1_R)$$

is the identity of S ;

3. if R is a field, then the transported structure on S is also a field.

5.3 Module

Theorem 5.3. (Transport of module structure and uniqueness)

Let R be a ring, let $(R, M, +, \mu)$ be an R -module, let N be a set, and let

$$\sigma : M \rightarrow N$$

be a bijection.

Define a binary operation $\oplus$ on N and a scalar action ν of R on N by

$$x \oplus y := \sigma(\sigma^{-1}(x) + \sigma^{-1}(y)), \quad \forall x, y \in N,$$

and

$$\nu(r, x) := \sigma(\mu(r, \sigma^{-1}(x))), \quad \forall r \in R, x \in N.$$

Then $(R, N, \oplus, \nu)$ is an R -module, and

$$\sigma : (R, M, +, \mu) \rightarrow (R, N, \oplus, \nu)$$

is an R -module isomorphism.

Moreover, $(\oplus, \nu)$ is the unique R -module structure on N for which σ is an R -module isomorphism.

Proof. We first prove existence.

Since $(M, +)$ is an abelian group and $\sigma : M \rightarrow N$ is a bijection, the group transport theorem shows that the operation

$$x \oplus y := \sigma(\sigma^{-1}(x) + \sigma^{-1}(y))$$

makes $(N, \oplus)$ into an abelian group, and that

$$\sigma : (M, +) \rightarrow (N, \oplus)$$

is a group isomorphism.

Now define a scalar action $\nu : R \times N \rightarrow N$ by

$$\nu(r, x) := \sigma(\mu(r, \sigma^{-1}(x))).$$

We prove that $(R, N, \oplus, \nu)$ is an R -module.

Take any $r, s \in R$ and $x, y \in N$.

(compatibility with addition in the ring) We have

$$\nu(r + s, x) = \sigma(\mu(r + s, \sigma^{-1}(x))).$$

Since $(R, M, +, \mu)$ is an R -module,

$$\mu(r + s, \sigma^{-1}(x)) = \mu(r, \sigma^{-1}(x)) + \mu(s, \sigma^{-1}(x)).$$

Therefore

$$\nu(r + s, x) = \sigma(\mu(r, \sigma^{-1}(x)) + \mu(s, \sigma^{-1}(x))).$$

Because σ is additive with respect to $+$ and $\oplus$,

$$\nu(r + s, x) = \sigma(\mu(r, \sigma^{-1}(x))) \oplus \sigma(\mu(s, \sigma^{-1}(x))) = \nu(r, x) \oplus \nu(s, x).$$

(compatibility with addition in the module) Similarly,

$$\nu(r, x \oplus y) = \sigma(\mu(r, \sigma^{-1}(x \oplus y))).$$

Since

$$\sigma^{-1}(x \oplus y) = \sigma^{-1}(x) + \sigma^{-1}(y),$$

we get

$$\nu(r, x \oplus y) = \sigma(\mu(r, \sigma^{-1}(x) + \sigma^{-1}(y))).$$

Using the module axiom in M ,

$$= \sigma(\mu(r, \sigma^{-1}(x)) + \mu(r, \sigma^{-1}(y))).$$

Hence

$$\nu(r, x \oplus y) = \nu(r, x) \oplus \nu(r, y).$$

(associativity of scalar action) We compute

$$\nu(r, \nu(s, x)) = \sigma(\mu(r, \sigma^{-1}(\nu(s, x)))).$$

By definition of ν ,

$$\nu(s, x) = \sigma(\mu(s, \sigma^{-1}(x))),$$

so

$$\sigma^{-1}(\nu(s, x)) = \mu(s, \sigma^{-1}(x)).$$

Therefore

$$\nu(r, \nu(s, x)) = \sigma(\mu(r, \mu(s, \sigma^{-1}(x)))).$$

Since M is an R -module,

$$\mu(r, \mu(s, \sigma^{-1}(x))) = \mu(rs, \sigma^{-1}(x)).$$

Hence

$$\nu(r, \nu(s, x)) = \sigma(\mu(rs, \sigma^{-1}(x))) = \nu(rs, x).$$

(identity axiom) Assume R has identity 1_R , as usual. Then

$$\nu(1_R, x) = \sigma(\mu(1_R, \sigma^{-1}(x))) = \sigma(\sigma^{-1}(x)) = x.$$

Thus $(R, N, \oplus, \nu)$ is an R -module.

Now we show that σ is an R -module isomorphism.

It is already a group isomorphism on the additive structures. It remains to check compatibility with scalar action. Take any $r \in R$ and $m \in M$. Then

$$\sigma(\mu(r, m)) = \nu(r, \sigma(m)),$$

because

$$\nu(r, \sigma(m)) = \sigma(\mu(r, \sigma^{-1}(\sigma(m)))) = \sigma(\mu(r, m)).$$

So σ is R -linear. Since σ is bijective, it is an R -module isomorphism.

This proves existence.

Now we prove uniqueness.

Suppose $(\oplus', \nu')$ is another R -module structure on N such that

$$\sigma : (R, M, +, \mu) \rightarrow (R, N, \oplus', \nu')$$

is an R -module isomorphism.

Since σ is then an isomorphism of additive groups, the uniqueness statement in the group transport theorem implies

$$\oplus' = \oplus.$$

It remains to show

$$\nu' = \nu.$$

Take any $r \in R$ and $x \in N$. Since σ is bijective, there exists a unique $m \in M$ such that

$$x = \sigma(m).$$

Because σ is R -linear with respect to ν' ,

$$\nu'(r, x) = \nu'(r, \sigma(m)) = \sigma(\mu(r, m)).$$

On the other hand, by definition of ν ,

$$\nu(r, x) = \sigma(\mu(r, \sigma^{-1}(x))) = \sigma(\mu(r, m)).$$

Hence

$$\nu'(r, x) = \nu(r, x).$$

Since $r \in R$ and $x \in N$ were arbitrary, it follows that

$$\nu' = \nu.$$

Therefore the R -module structure on N is unique. □